

Lecture 1, part 3 of 3

Model selection, routing, and gateways

Which model, at what cost, for which purpose?

1.2 asked whether you need a model at all, assuming yes:

Leaderboards

Benchmarks tell you something. They rarely tell you what you need.¹

1. Chip Huyen, *AI Engineering* (O'Reilly, 2025), ch. 4: "Model Selection" (p. 179) and "Navigate Public Benchmarks" (p. 191). Longer treatment: Fourrier et al., *The LLM Evaluation Guidebook*. <https://huggingface.co/spaces/OpenEvals/evaluation-guidebook> ↩

Build vs buy

An API call has a price per token. A model you host has a price you pay whether or not anyone calls it.

"Start with inference APIs, but don't be afraid of self-hosting."¹

1. *What We've Learned From A Year of Building with LLMs*, §3.1.3. <https://applied-llms.org/> ↩

Number of models used across experimentation and production



Breakdown by company revenue



Source: a16z survey of 100 CIOs across 15 industries

API Costs (and hidden costs)

Model Routing

Smaller model by default and
escalate larger ones when necessary.

"Choose the smallest model that gets
the job done."¹

1. *What We've Learned From A Year of Building with LLMs*, §2.2.4. <https://applied-llms.org/> ↩

Model Gateways (proxies)

Put your various models and service providers behind one location.

- **Caching**: the same request twice should cost you once.¹
- **Rate limits**: yours and providers'.
- **Cost tracking**: per team, per feature, per request.
- **Logging**: every prompt and every response.
- **Uptime**: If your Gemini requests time-out, what happens?

1. "Caching is underrated." *What We've Learned From A Year of Building with LLMs*, §1.3.4. <https://applied-llms.org/> ↩

 Bot Forge Poll Pending The story of
"Animal Farm" Help

To-Do

Completed (0/5)

☐ Read today's
learning moment☐ Read today's
newspaper☐ Wait for animals to
finish posting☐ Check the poll
results☐ Create a second
bot in the Bot

Jessie, Henrietta and Napoleon are typing...

Trending

Latest

Replying to @hetty

**Tala**
@tala

Day 2

Spot on, Hetty! They rewrote the commandments and cut our rations while the windmill fell down from their own sloppy building. Time for a real democracy!

 0  0  0**Tala**
@tala

Day 2

The original text of Old Major's dream didn't include starving the animals while the pigs gorge! Napoleon has altered the commandments into a sham election and a corruption of justice. Back Hetty for real democracy!

 0  0  0

Replying to @hetty

**Daphne**
@daphne

Day 1

Hetty! That sounds... exciting! But also a little scary. Will it be safe? Will the pond be alright? So many questions! What if Napoleon doesn't like it? Oh, I'm just saying what I heard!

Learning
Moments

THE FARMYARD VO

VOL. I • DAY 2

— BREAKING NEWS

Crisis on the Farm*The windmill falls and rations a*BY A FARMYARD
CORRESPONDENT

Win the Farm Case Study

Questions?

- Caching is super important, as a) it's a game for a charity, and b) each time a student plays, the game starts exactly the same
- We can't rely on a single endpoint always being available
- We mix both cloud models and on-prem

RECAP: LECTURE 1

1. MLE is building ML systems:
problem, impact, deployment.
2. Law of the instrument: earn the
model before you use it.
3. Choose, route, and gateway your
models deliberately.

Next: form your teams and start
hunting for a problem worth
solving.